

QUIZ III - Z-TRANSFORMS & BODE DIAGRAMS

1. A discrete-time system has impulse response

$$h[n] = \delta[n] + (1/2)^n u[n] + (-5)^n u[n].$$

What is the region of convergence of $H(z)$?

(A) $|z| > 5$

(B) $Re(z) > 5$

(C) $|z| < 1$

	Signal, $x(n)$	z -Transform, $X(z)$	ROC
1	$\delta(n)$	1	All z
2	$u(n)$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
3	$a^n u(n)$	$\frac{1}{1 - az^{-1}}$	$ z > a $
4	$na^n u(n)$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $
5	$-a^n u(-n - 1)$	$\frac{1}{1 - az^{-1}}$	$ z < a $
6	$-na^n u(-n - 1)$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z < a $
7	$(\cos \omega_0 n) u(n)$	$\frac{1 - z^{-1} \cos \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
8	$(\sin \omega_0 n) u(n)$	$\frac{z^{-1} \sin \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
9	$(a^n \cos \omega_0 n) u(n)$	$\frac{1 - az^{-1} \cos \omega_0}{1 - 2az^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $
10	$(a^n \sin \omega_0 n) u(n)$	$\frac{az^{-1} \sin \omega_0}{1 - 2az^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $

$$H(z) = 1 + \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 + 5z^{-1}}$$

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2. A system has transfer function

$$H(z) = \frac{(1 - \frac{1}{4}z^{-1})}{(1 - \frac{1}{2}z^{-1})(1 + 3z^{-1})}$$

Is the system stable? **$z = \frac{1}{2}$** **$z = -3$**

(A) Yes

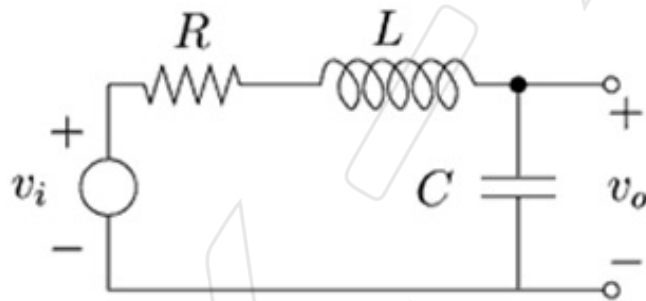
Ⓑ No

(C) It is stable for $|z| > 3$.

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3. A circuit has transfer function

$$H(s) = \frac{1}{s^2 LC + sRC + 1}.$$



What is the (possible) resonant frequency of the circuit?

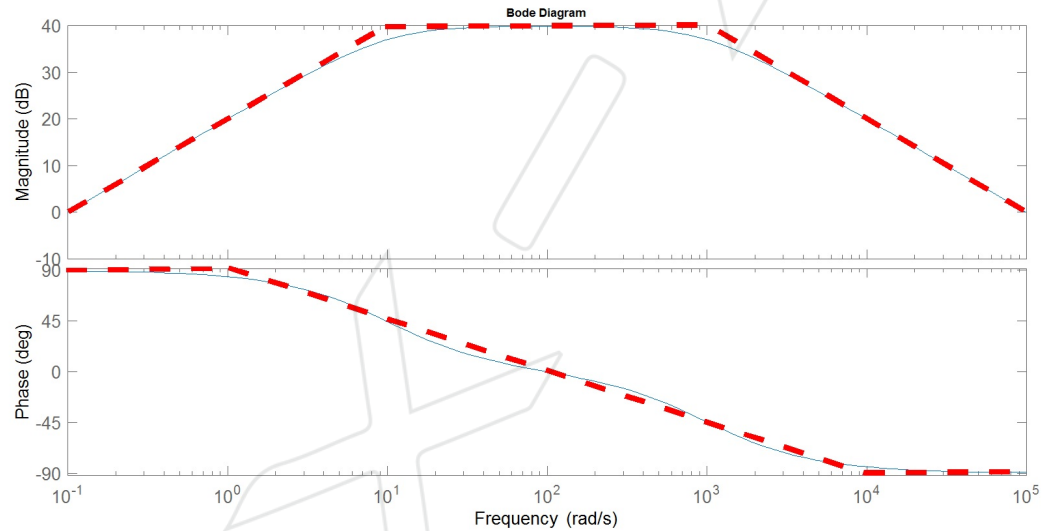
(A) $\omega_0 = \sqrt{LC}$

Ⓑ $\omega_0 = \frac{1}{\sqrt{LC}}$

(C) $\omega_0 = R\sqrt{\frac{C}{L}}$

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4. What is the transfer function of the system with Bode diagram



(A) $H(s) = \frac{10^2(s+1)}{(s+10)(s+1000)}$

(B) $H(s) = \frac{10^3}{s(s+10)(s+1000)}$

Ⓒ $H(s) = \frac{10^5 s}{(s+10)(s+1000)}$

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1. A discrete-time system has impulse response

$$h[n] = \delta[n] + (1/2)^n u[n] + (-5)^n u[n].$$

What is the region of convergence of $H(z)$?

(A) $|z| > 5$

(B) $\text{Re}(z) > 5$

(C) $|z| < 1$

Solution :

$$H(z) = \mathcal{Z}\{h[n]\} = \mathcal{Z}\{\delta[n]\} + \mathcal{Z}\left\{\left(\frac{1}{2}\right)^n u[n]\right\} + \mathcal{Z}\{(-5)^n u[n]\}$$

$$= 1 + \frac{1}{1 - \frac{1}{2} z^{-1}} + \frac{1}{1 + 5 z^{-1}}$$

(for all z) (for $|z| > \frac{1}{2}$) (for $|z| > 5$)

most "restrictive" 

$\Rightarrow \text{ROC of } H(z): |z| > 5$

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2. A system has transfer function

$$H(z) = \frac{(1 - \frac{1}{4}z^{-1})}{(1 - \frac{1}{2}z^{-1})(1 + 3z^{-1})}$$

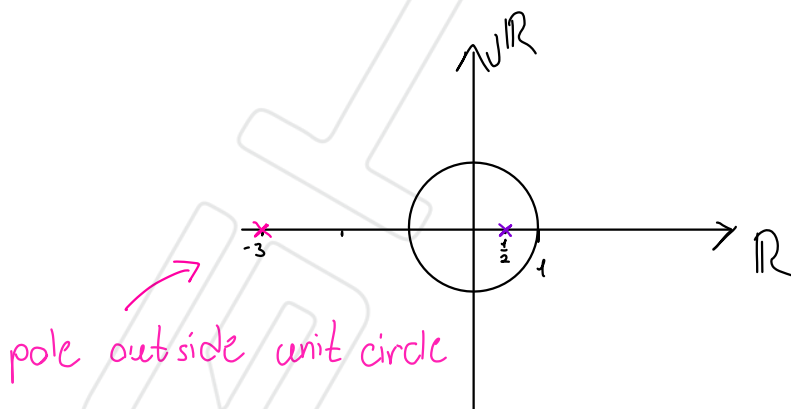
Is the system stable?

(A) Yes

(B) No

(C) It is stable for $|z| > 3$.

Solution: poles of $H(z)$: $z = \frac{1}{2}$ and $z = -3$

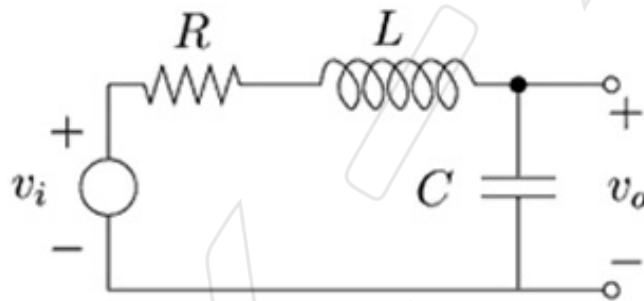


$\Rightarrow H(z)$ is unstable

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3. A circuit has transfer function

$$H(s) = \frac{1}{s^2 LC + sRC + 1}.$$



What is the (possible) resonant frequency of the circuit?

(A) $\omega_0 = \sqrt{LC}$

(B) $\omega_0 = \frac{1}{\sqrt{LC}}$

(C) $\omega_0 = R\sqrt{\frac{C}{L}}$

Solution: (also already given slides for Lectures 12-13)

compare

$$s^2 LC + s RC + 1 \longleftrightarrow s^2 T^2 + s(2\xi T) + 1$$

$$LC = T^2 \rightarrow T = \sqrt{LC}$$

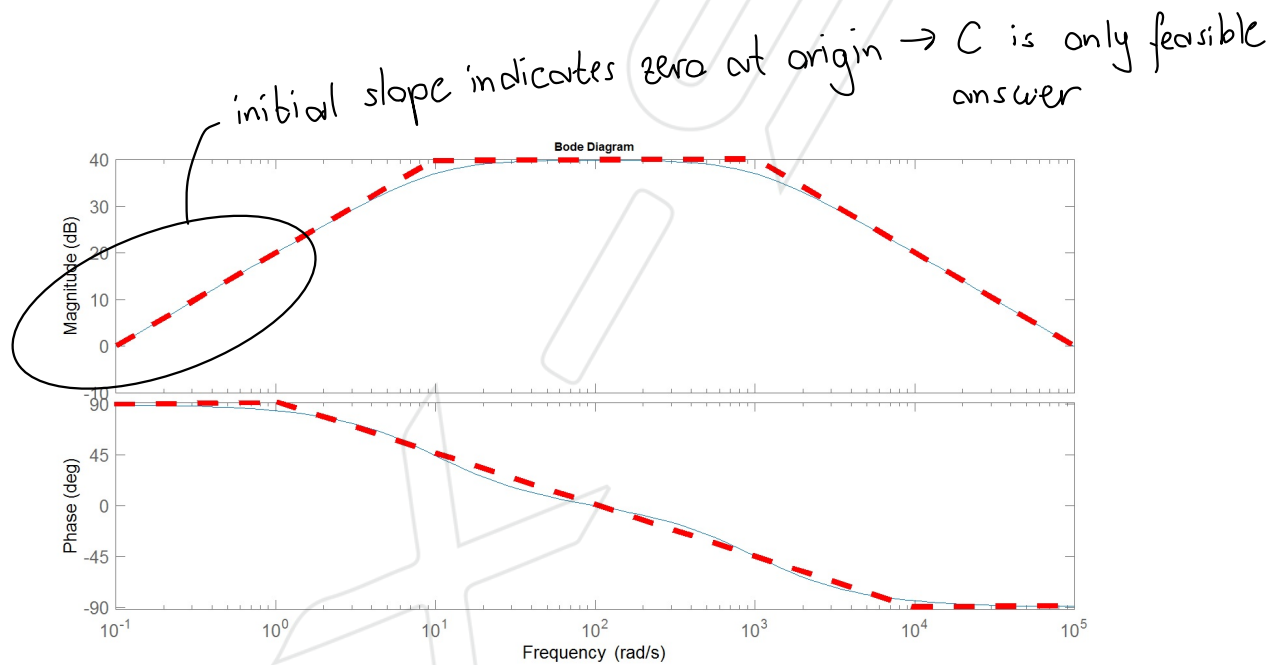
$$RC = 2\xi T \rightarrow \xi = \frac{RC}{2T} = \frac{1}{2} R \sqrt{\frac{C}{L}}$$

The resonance frequency (also called natural frequency) is

$$\omega_0 = \frac{1}{T} = \frac{1}{\sqrt{LC}}$$

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4. What is the transfer function of the system with Bode diagram

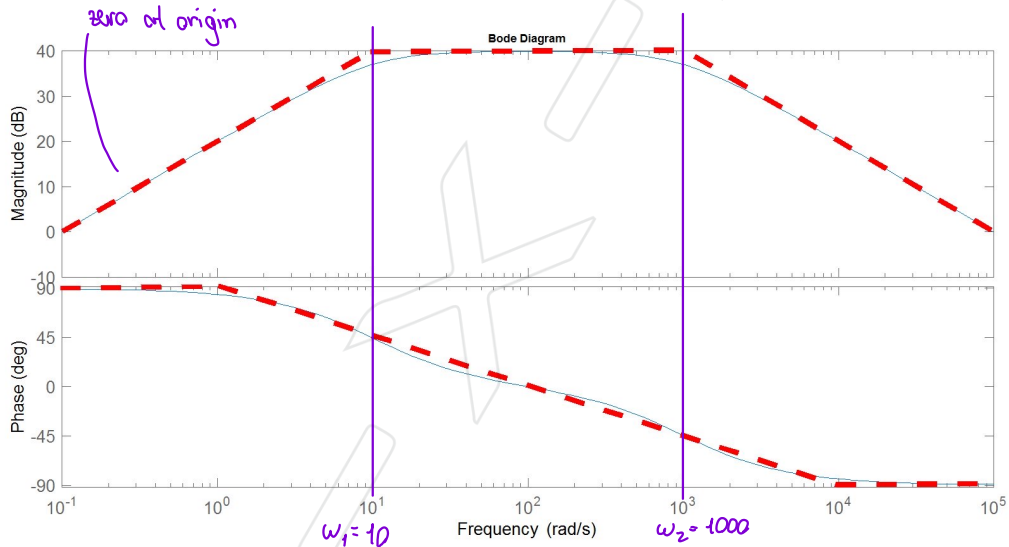


(A) $H(s) = \frac{10^2(s+1)}{(s+10)(s+1000)}$

(B) $H(s) = \frac{10^3}{s(s+10)(s+1000)}$

(C) $H(s) = \frac{10^5 s}{(s+10)(s+1000)}$

However, it is a good exercise to determine $H(s)$ from the diagram:



at / around $\omega_1 = 10$: slope of $|H(j\omega)|$ changes by -20 dB/dec

$\angle H(j\omega)$ changes by -90° between $\omega = 10^0$ and $\omega = 10^2$

↳ LHP pole: $s = -10$

at / around $\omega_2 = 10^3$ slope of $|H(j\omega)|$ changes by -20 dB/dec

$\angle H(j\omega)$ changes by -90° between $\omega = 10^2$ and $\omega = 10^4$

↳ LHP pole: $s = -1000$

$$H(s) = K \frac{s}{\left(1 + \frac{1}{10}s\right)\left(1 + \frac{1}{1000}s\right)}$$

$$= 10 \frac{s}{\left(1 + \frac{1}{10}s\right)\left(1 + \frac{1}{1000}s\right)}$$

$$H(s) = \frac{10^5 s}{(s + 10)(s + 1000)}$$

find K : $|H(j\omega)| \Big|_{\omega=0.1} = 0 \text{ dB} = 1$

$$1 = \left| K \frac{0.1j}{\underbrace{\left(1 + \frac{1}{10}0.1j\right)\left(1 + \frac{1}{1000}0.1j\right)}_{1 + 0.01j + 0.0001j - 10^{-6}}} \right| \approx |K \cdot 0.1j|$$

$$\Rightarrow K = 10$$